

Team 4 Presentation

Currency Converter Application "XChange"

Team Members & Roles:

Krish – Frontend

Anastasiia – Database & Persistence

Selina – Middle Tier / Business Logic

Project overview

- A full stack web application that enables you to exchange between currencies from the top 25 most used currencies in the world.
- You will also be able to favorite pairs of currencies that you frequently exchange to and from.
- The application also allows you to save your conversion history and add in a little note if you want to see it later.

Main Features

- User registration and login
- Currency converter
- Live exchange rates via external API
- User-specific favourite currency pairs
- Quick conversion using favourite pairs
- Swap currencies
- User-specific conversion history
- Add notes to conversions
- Currency list with search
- Delete confirmation dialogs

Stretch Goals:

- Travel budget tracking + timestamps
- Advanced search and filtering
- Edit notes

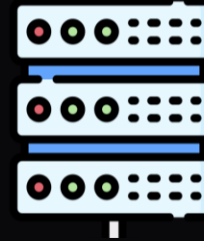
Technologies used



Frontend

React, Vite and Tailwind CSS

Built a responsive user interface



Backend

Java 17, Spring Boot

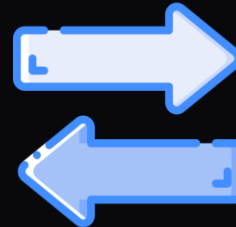
Implemented business logic and REST API endpoints



Database

MySQL, JdbcTemplate

Stored user information, favourite currency pairs and conversion history



External API

UniRate

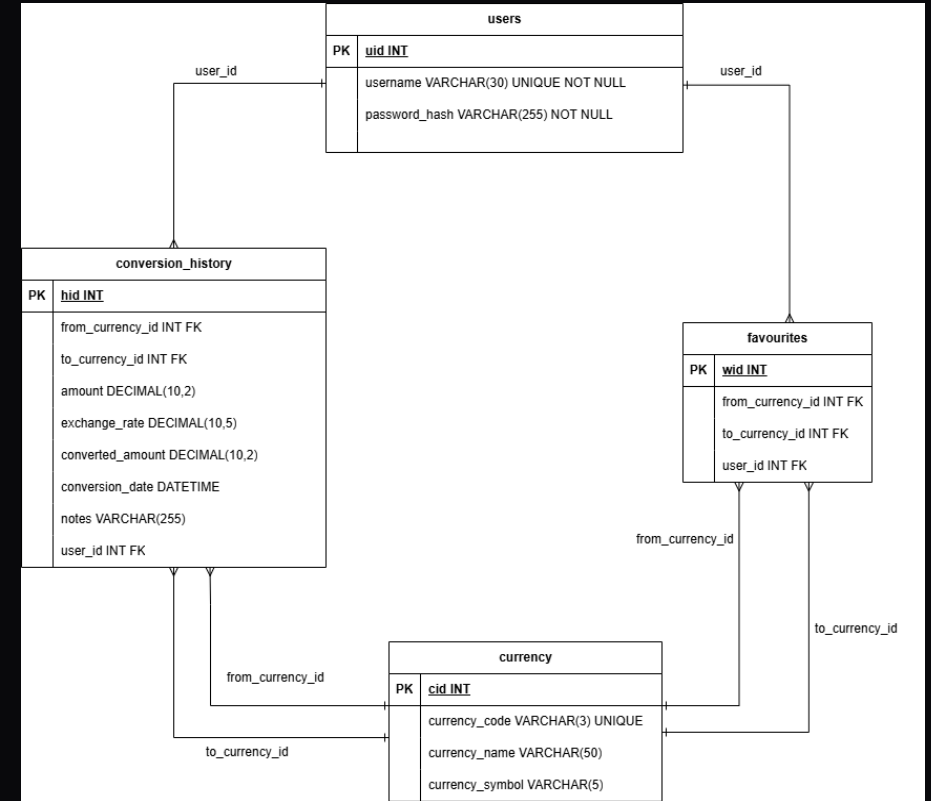
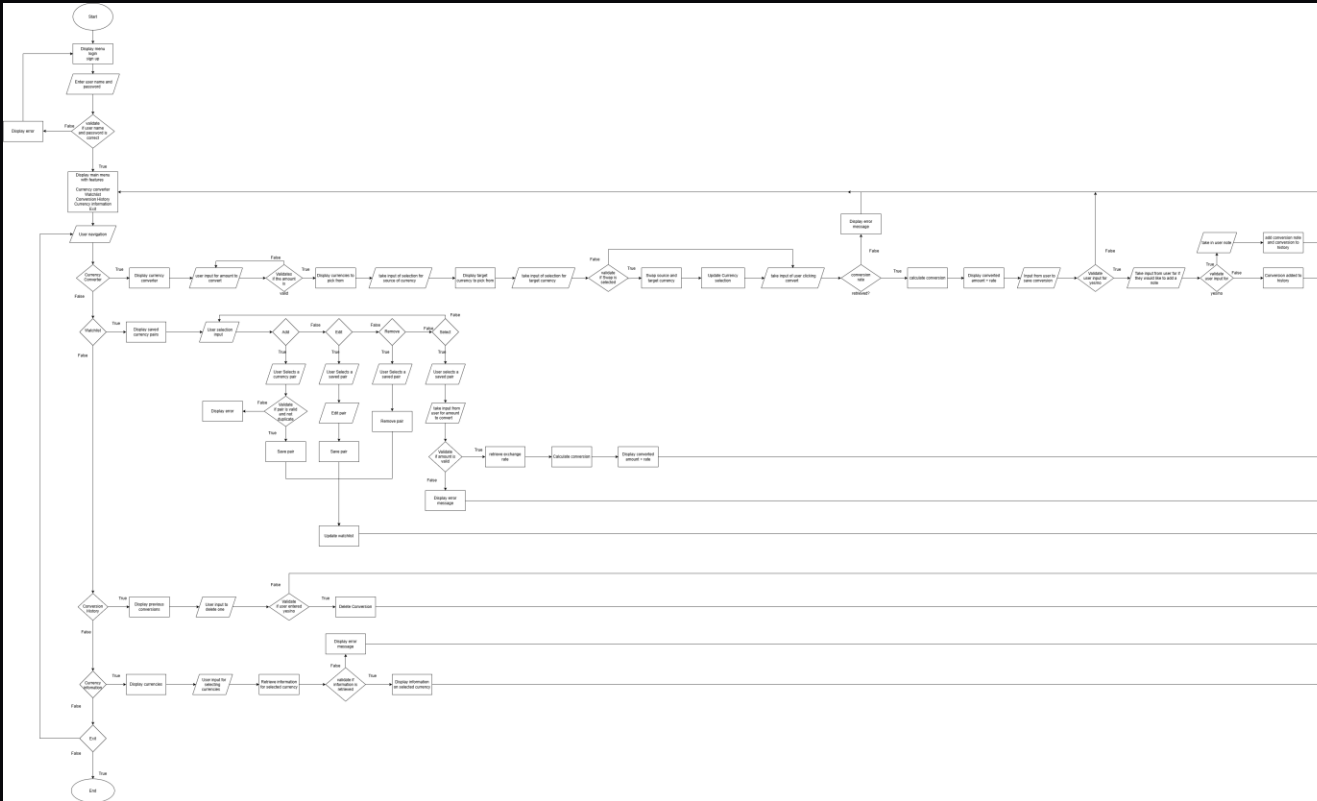
Provides live currency exchange rates



Development & testing Tools

IntelliJ IDEA, Postman, Git & GitHub

Flowchart / ERD



Links: [Flowchart](#) / [ERD](#)

Live demo

Testing and Validation

Backend Testing

- JUnit 5 tests for DAO operations.
- Tested database CRUD functionality: create, read, update and delete.
- Used a separate test database to avoid affecting application data.

API & Integration Testing

- Tested REST endpoints using Postman.
- Verified integration with the UniRate currency exchange API.
- Checked data retrieval and conversion functionality.

Frontend Validation

- Validated user input for currency conversions.
- Added search functionality to filter currencies by name or code.
- Checked navigation and functionality across application pages.

Challenges and Solutions

Challenge	Solution
Database design	Created a relational database with foreign keys to connect users, currencies, favourites and conversion history.
Duplicate favourites	Added database constraints and backend error handling to prevent duplicate currency pairs.
Team collaboration	Used Git branches to develop features independently and merged changes into the main branch.
Frontend and backend integration	Connected React components to Spring Boot REST endpoints and verified requests and responses using Postman.

Version 2.0 Features

1. Users could define a target exchange rate for a currency pair. The application would monitor the rate and notify the user when the target is reached.

For example:

GBP → EUR

Example current rate

1.1464



Target rate

1.1800

Notify when GBP/EUR reaches or exceeds 1.1800

2. Users could view how a currency pair's exchange rate has changed over a selected period, such as one week, one month or one year.

Q&A

